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|----------|----------------------------------------------------------------------------------------------------------------|-----------|
|          |                                                                                                                |           |
| 3(a)(i)  | mention of upthrust and weight                                                                                 | <b>B1</b> |
| 3(a)(ii) | upthrust is greater than the weight                                                                            | <b>B1</b> |
|          | (resultant force is) upwards                                                                                   | <b>B1</b> |
| 3(b)     | $A$ , $\rho$ , $g$ and $M$ are constant                                                                        | <b>B1</b> |
|          | <i>either</i> acceleration $\propto$ – displacement                                                            | <b>B1</b> |
|          | <i>or</i> acceleration $\propto$ displacement <u>and</u> (– sign indicates) $a$ and $x$ in opposite directions |           |
| 3(c)(i)  | <i>either</i> $\omega = 2\pi / T$ <i>or</i> $\omega = 2\pi f$ <u>and</u> $f = 1 / T$                           | <b>C1</b> |
|          | $\omega = 2\pi / 1.3$<br>$= 4.8 \text{ rad s}^{-1}$                                                            | <b>A1</b> |
| 3(c)(ii) | $\omega^2 = A\rho g / m$                                                                                       | <b>C1</b> |
|          | $4.83^2 = (4.5 \times 10^{-4} \times \rho \times 9.81) / 0.17$                                                 | <b>C1</b> |
|          | $\rho = 900 \text{ kg m}^{-3}$                                                                                 | <b>A1</b> |